

TENUGHAT VIDYUT NIGAM LIMITED

(A GOVT. OF JHARKHAND UNDERTAKING)

तेनुघाट विद्युत निगम लिमिटेड

Website: - tvnl.in

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NIT No. 30/BMD/W/TVNL/RAN/2026-27

SRM RFx No-1000014024



TENDER DOCUMENT

SUBJECT:-(1) ANNUAL RATE CONTRACT FOR BOILER PRESSURE PARTS & H.P. VALVES OF BOILER OF UNIT NO. 1 & 2 AT TTPS, LALPANIA for the period of two years.

AND

(2) ANNUAL RATE CONTRACT & OVERHAULING WORKS FOR DUCTS/DAMPERS/GATES OF BOILER OF UNIT NO. 1 & 2 AT TTPS, LALPANIA for the period of two years.



TENUGHAT VIDYUT NIGAM LIMITED

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(A GOVT. OF JHARKHAND UNDERTAKING)

JUPMI Building premises in ABD Area, Ranchi Smart City, P.O.&P.S. Dhurwa, Ranchi-04, (Jharkhand)

Ref No.686/2026-27

Dated 19.08.2026

NOTICE INVITING TENDER

NIT No. 30/BMD/W/TVNL/RAN/2026-27, SRM RFx No-1000014024

On behalf of Tenughat Vidyut Nigam Limited (TVNL), Ranchi online tender is invited (in two parts Part-I - Technical & Commercial Bid, Part-II - Price Bid to be submitted in TVNL's website <https://webdispatcher.tvnl.in/srmprd> only) from experienced, reputed and financially sound firms for 1. Annual Rate Contract for Boiler Pressure Parts & H.P. Valves of Boiler of Unit No. 1 & 2 at TTPS, Lalpania for the period of two years. 2. Annual Rate Contract for APH, Scanner Fan, Dampers and Gates of Boiler of Unit No. 1 & 2 for the period of two years. Requisite cost of Tender Document & Earnest Money is to be submitted to designated account TVNL well before submission deadline. Hard copy of Part-I-Technical & Commercial bid is also to be submitted in TVNL Headquarter JUPMI Building Premises in ABD, Ranchi Smart City, P.O.- Dhurwa- 834004, Jharkhand on or before due date. The detail of Tender follows:

Description of Work	1. Annual Rate Contract for Boiler Pressure Parts & H.P. Valves of Boiler of Unit No. 1 & 2 at TTPS, Lalpania for the period of two years. 2. Annual Rate Contract for APH, Scanner Fan, Dampers and Gates of Boiler of Unit No. 1 & 2 for the period of two years.
Estimated Cost	Rs. 1,12,05,126.11/- (Rupees One crore twelve lakh five thousand one hundred twenty-six and eleven paise) only plus GST as applicable.
Cost of Tender Documents (Non-Refundable) to be deposited through payment gateway link in TVNL website	Tender fee Rs. 15,000/- (Rupees fifteen thousand) only. GST@18% Rs. 2,700/- Total Tender fee Rs. 17,700/- (Rupees seventeen thousand seven hundred) only. Inclusive GST charges.
Amount of EM to be deposited in the form of BG/through payment gateway	Rs. 2,24,110/- (Rupees Two lakh twenty-four thousand one hundred ten) only.
Bank details for BG creation	Bank Name: - ICICI Bank Ltd, Name of Account: - Tenughat Vidyut Nigam Ltd. Branch Name: - Lalpur, branch IFSC Code: - ICIC0000292
Downloading of Tender Paper	From 24/08/2026
Last Date& Time for Purchase of Tender Paper	11/09/2026 up to 14:00 Hrs.
Last Date & Time of Bid Submission	14/09/2026 up to 14:00 Hrs.
Date & Time of Opening of Part-I of Tender	14/09/2026 at 16:00 Hrs.
Place of submission of bid documents and address for communication	Electrical Superintending Engineer (C&M), Tenughat Vidyut Nigam Limited, JUPMI Building Premises in ABD, Ranchi Smart City, P.O.- Dhurwa, Ranchi- 834004, Jharkhand

1.The Bidder should have the following Qualifying Requirements (QR):

TECHNICAL QUALIFYING CRITERIA: -

(a) The bidders should have successfully executed experience of carrying out OH/ Maintenance work of Boiler pressure parts OR OH/Maintenance work of H.P. Valves of Boiler OR OH/ Maintenance work of Non pressure parts (APH / Scanner Fan/Damper Gates) in any Thermal power plant not less than 210 MW unit capacity of any Govt. / Semi Govt. /CPSU/SPSU/ SEB's/ Reputed Private Organization.

(b) The bidder should have experience of having successfully completed similar works along with successfully completion certificate during last 5(five) years ending last day of month previous to the one in which tender/NIT is invited. Bidder has to submit copy of work order along with its successful completion certificate. The successful completed similar work shall be either of the following: -

- i. Three similar completed works each costing not less than the amount equal to 40% of estimated cost.

OR

- ii. Two similar completed works each costing not less than the amount equal to 50% of estimated cost.

OR

- iii. One similar completed works costing not less than the amount equal to 80% of estimated cost.

Similar work means the bidder must have successfully executed experience of carrying out OH/ Maintenance work of Boiler pressure parts OR OH/Maintenance work of H.P. Valves of Boiler OR OH/ Maintenance work of non pressure parts (APH / Scanner Fan/Damper Gates) in any Thermal power plant not less than 210 MW unit capacity of any Govt. / Semi Govt. /CPSU/SPSU/ SEB's/ Reputed Private Organization

(c) The firm has to submit a self-declaration undertaking that their company has not been blacklisted by NTPC/BHEL/SEBs/any PSU / Any Thermal unit at the time of submission of bid in NIT.

(d) The bidder should have a valid GST Registration Certificate, Employee Provident Fund (EPF) Code Number, Employees' State Insurance (ESI) registration, and Ownership/Partnership Deed, as applicable.

FINANCIAL QUALIFYING CRITERIA: -

- I. Average annual turnover of preceding 03 (three) years (i.e. 2022-23, 2023-24 & 2024-25) should not be less than 30% of the estimated cost. Annual Accounts Reports of preceding 03 years certified by Auditor/CA for determining, overall profitability, average annual turnover etc, are to be submitted.
In case where audited result for the last preceding financial year are not available, certificate of financial statements from a practicing Chartered Accountant shall also be considered.
 - II. Solvency certificate should be for current FY or FY 2024-25. The bidder should have financial credibility/capacity of at least 10% of the estimated cost. Bidder may submit Bank Solvency Certificate of any Indian National/Scheduled Commercial Bank in support of their financial credibility/capacity.
 - III. The bidder should have an active GST number. Copies of the GSTR-3B return and the latest Income Tax Return (ITR) shall be submitted
2. The tender document is to be downloaded from the TVNL website www.tvnl.in. Tenders will have to be submitted in TVNL Web Portal only website <https://webdispatcher.tvnl.in/srmpd>

For easy reference “**Vendor Bidding Manual for Open Tender**” can be downloaded from the TVNL website homepage.

3. After advertisement of NIT, intended bidder will first communicate with TVNL contact person Shri Kavindra Ram, EEE(C&M) for this aforesaid NIT only on his Mobile No-9835127953 and send their firm details like Name, address, email, phone nos. and brief company’s credentials well before the submission deadline. Also company’s contact person name, email, Mobile no may be sent in email ID: k.ram@tvnl.in, so that login ID & password can be created in SRM-SAP which will be sent to bidders contact person email. Bidder contact person has to register at TVNL website: <https://webdispatcher.tvnl.in/srmprd> with the given ID & password.
4. The cost of tender document shall be deposited well before the submission deadline of the tender in the shape of Bank Draft issued by any Nationalized/Scheduled Bank in favour of “**TENUGHAT VIDYUT NIGAM LIMITED**” payable at Ranchi which may be submitted to TVNL contact person, in person or communicate to contact person by email by attaching the scanned copy of the Demand draft so that the contact person will allow access to open the RFx details and initiate bidding at TVNL website and attach the details of Part-I & fill the details of Part- II.
5. Bidder have to fill as well as attach scanned copies of the related documents to qualify in the part-I (Technical & commercial) on TVNL website: <https://webdispatcher.tvnl.in/srmprd> and also the bidder has to submit the hardcopy of bid part-I (Technical & commercial) to the ESE (C&M), TVNL, Ranchi office before the submission deadline along with the cost of tender document and BG of Earnest money deposit (EMD).
6. Bidders have to bid for part-II (Price part) on TVNL website: <https://webdispatcher.tvnl.in/srmprd> only. **Price Part is not to be submitted in Hard copy** (in sealed envelope).
7. Bidder will not get access to bidding in SRM-SAP without providing the details of cost of tender document. Any type of tempering/alteration with tender document shall not be allowed and offer, if submitted so, will be summarily rejected.
8. Tender without Earnest Money shall not be considered. Offers shall be accompanied with the Cost of Tender document and Earnest Money Deposit (EMD) of requisite value as specified above in a separate envelope failing which offers are liable for rejection. EMD may be furnished in the form of Demand Draft/ Bank Guarantee in required format from any Indian Nationalized Bank/ Schedule Commercial Bank in favour of “TENUGHAT VIDYUT NIGAM LIMITED” payable at Ranchi. The Bank Guarantee should remain valid for 12 months.
9. The Earnest Money will be forfeited on revocation of tender before expiry of validity of the tender or on refusal to enter into the contract after the award is made by TVNL to the bidder.
10. Any corrigendum/Addendum/Extension of bid opening date and cancellation of NIT will be published through our website www.tvnl.in only and not through newspaper. The bidders are requested to visit our website regularly for any addendum/corrigendum/extension till opening of NIT.
11. The due date of bid opening as mentioned in the tender will remain firm. However, in case of unforeseen circumstances or due to administrative reasons the bids are not opened on the due date, the same shall be opened on the next working day at the same time. In case the last date of submission of bids is declared as holiday, the bids received up to the time of closing of bids on the next working day shall be considered for opening.
12. TVNL reserves the right to reject any / all offer increase/decrease or to distribute the quantity among the bidders without assigning any reason thereof and not bound to accept the lowest tender or to assign any reason for the rejection of the lowest tender.

(R. Ranjan)

Electrical Superintending Engineer (C&M)



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JUPMI Building premises in ABD Area, Ranchi Smart City, P.O&P.S.Dhurwa, Ranchi-04, (Jharkhand)

BILL OF QUANTITY

PART- I (TECHNICAL& COMMERCIAL)

Sub: - 1. Annual Rate Contract for Boiler Pressure Parts & H.P. Valves of Boiler of Unit No. 1 & 2 at TTPS, Lalpania for the period of two years. 2. Annual Rate Contract for APH, Scanner Fan, Dampers and Gates of Boiler of Unit No. 1 & 2 for the period of two years.

Sl. No.	Particulars	Details to be furnished by the Bidder (Enclose separate sheets if required)	Flag No. It is mandatory	Page No.
1.	Name of the Bidder (Enclose Letter Head of firm)		I	
2.	Address of Bidder with Telephone, Fax, E-mail		II	
3.	Name of the Proprietor/ Head of the Firm and its key personnel with organizational chart giving details of qualification and experience etc.		III	
4.	Details of maintenance set up with resources like permanent employees (with designation & strength), tools & tackle and other equipment with capacities		IV	
5.	Name and Qualification of the site-in-charge		V	
6.	Work Experience of Bidder (Attach copy of work order of similar work and its successful completion certificate during last seven years as per S. N. 1. e. of QR)	Work Order No..... Dated..... Completion Certificate Dated..... Work Order No..... Dated..... Completion Certificate Dated..... Work Order No..... Dated..... Completion Certificate Dated.....	VI	
7	The bidder should have an active GST number. Copies of the GSTR-3B return and the latest Income Tax Return (ITR) shall be submitted		VII	
8.	Financial Credibility/ Capacity (Attach latest Bank Solvency Certificate)		VIII	

9.	Average annual turnover of bidder during the last 03 financial years (Attach audited Balance Sheet & P/L Account for last three years ending 31 st March, 2025)	Average annual turnover of bidder during the last 03 (three) financial years Rs..... Annual Turnover FY 2022-23: Rs..... FY 2023-24: Rs..... FY 2024-25: Rs.....	IX	
10.	Name & Designation of Authorized Signatory (Attach Power of Attorney if applicable)		X	
11.	Nature of Firm (Proprietorship/Partnership /Pvt. Ltd./Public Sector) (Attach proof)		XI	
12.	Permanent Account Number (PAN) (Attach proof)		XII	
13.	GST Registration Number (Attach proof)		XIII	
14.	EPF. Code No. (Attach copy of registration)		XIV	
15.	Particulars of Earnest Money (Transaction ID No. /BG No. & Date)		XV	
16.	Particulars of Cost of Tender Document (BOQ) (Transaction Id. & Date)		XVI	
17.	Whether the firm are a Govt. Organ. /Govt. U/T.		XVII	
18.	Whether the firm is registered with NSIC/SSI (Jharkhand). (enclose copy of certificate)		XVIII	
19.	Specification of the material to be supplied:		XIX	
20.	Quantity both in figure & in words:		XX	
21.	F.O.R (Prices Basis)		XXI	
22.	Packing & Forwarding charges		XXII	
23.	Freight		XXIII	
24.	GST (kindly indicate the taxes applicable)		XXIV	
25.	Payment Terms for materials		XXV	
26.	Transit Insurance charges	The TVNL have its own marine policy with insurance company.	XXVI	
27.	Transportation charges		XXVII	

28.	Validity period of the offer (minimum 120 days)		XXIII	
29.	Guarantee period of the material to be supplied		XXIX	
30.	Whether you will deposit Security Money. If not reasons thereof with documentary proof of exemption:		XXX	
31.	Whether you will furnish the Performance Guarantee		XXXI	
32.	Inspection		XXXII	
33.	Do you fulfill the pre- requisite qualification?		XXXIII	
34.	Whether you accept all term & conditions of B.O.Q.		XXXIV	
35.	Penalty		XXXV	
36.	Details required for ERP Vendor Coding:- a. Name of firm : b. Address : PIN: c. E-mail ID : d. Phone No. : e. Name of Contact Person: f. GST No. : g. PAN No : Details required for EMD Refund through RTGS/NEFT : h. Bank Name : i. Branch : j. Branch Code : k. A/c. No. : l. IFS Code : m. MICR Code :		XXXVI	
37.	Other details and remarks, if any		XXXVII	

(Signature & Seal of Bidder)

NOTE: - DETAILS SHEET/DOCUMENTS SUBMITTED ALONG WITH BID SHOULD BE FLAGGED. IT IS TREATED AS NECESSARY FOR SUBMISSION OF TENDER DOCUMENTS.



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(A GOVT. OF JHARKHAND UNDERTAKING)

JUPMI Building premises in ABD Area, Ranchi Smart City, P.O&P.S. Dhurwa, Ranchi-04, (Jharkhand)

INSTRUCTIONS TO BIDDERS FOR ONLINE BIDDING

Each of these instructions must be read carefully and followed strictly. Failure in complying with any of these would render your offer liable for non-submission and TVNL or its any department will not take any responsibility for any bid not successfully Submitted/Participated in **RFx** (Request **for Quotation**).

IMPORTANT STEPS FOR PARTICIPATION IN BIDDING:

If Bidder is interested to participate in the Open tender, then follow the below steps to provide the bid quotation.

Note: Use Only Internet Explorer browser to login into TVNL Portal.

- 1) Open RFx
- 2) Pay tender Fee & Earnest Money Deposit Amount and inform the same to TVNL contact person.
- 3) Click on participate button.
- 4) Click on Create Response Button.
- 5) Enter the Price in the Item tab.
- 6) Click on Technical RFx Button to upload the documents.
- 7) Finally click on Submit button before the Submission deadline date and time.

PAY THE TENDER FEE & EARNEST MONEY DEPOSIT (EMD) AMOUNT

Tender Fee & Earnest Money Deposit may be submitted to TVNL contact person in person or communicate to the contact person by email by attaching the scanned copy of the Demand draft/BG so that the contact person will allow access to open the RFx details and initiate bidding at TVNL website and attach the details of Part-I & fill the details of Part- II.

Inform TVNL contact person to provide the access to RFx to participate in the Bidding.

OPEN THE RFx NUMBER IN THE VENDOR PORTAL

Vendor login: Login to the TVNL Vendor Portal through the URL mentioned below.

Vendor Login URL: <http://webdispatcher.tvnl.in/srmpdr>

Login Page shall appear on Vendor Portal page.

Step-1: Enter the User ID and Password and Change the Password during the first login.

Step-2: Click on Logon button

Step-3: Click on **RFx and Auctions**.

Click on Event Number to open the required RFx No. For E.g. 1000000XXX

If Bidder is accessing the bidder portal for the first time then system will show the Register button once you have opened the RFx as shown in the below screen.

For the second time onwards system will not ask for bidder to Register.

After click on Register button, if bidder is not paid the “**Tender fee amount**” to TVNL then system will show the error message as shown in the below screen.

If you get the error message follow the below step:

- 1) First vendor will pay “**Tender fee & Earnest Money Deposit**” amount and inform the same to TVNL contractor to provide the access to RFx.

Note: Once contact person provide the access to RFx, then bidder will refresh button of the RFx and proceed further.

After the bidder Register, system will show the below options to **Participate, Do Not Participate, Tentative and Create Response**.

What is “Participate” button?

- If bidder click on participate that means, Bidder interested in participating the bidding.

What is “Do Not Participate” ?

- If bidder click on Do Not Participate that means, Bidder is not interested to participate in the bidding.

What is “Tentative” button?

- If bidder click on Tentative that means, Bidder either will participate or not participate in the bidding.

What is “Create Response” button?

- After click on participate/Tentative button, bidder will use the create response to provide his/her bid quotation.

TO CREATE THE RESPONSE FOR THE PARTICULAR RFx

Step-1: Open the RFx.

Step-2: Click on Create Response.

ANSWER THE QUESTIONS IN THE RFx

After clicking on create Response button, system will show the error message. At this time, bidder will follow below steps:-

- 1) Click on Questions tab in the screen
- 2) Bidder will find Questions
- 3) Answer to those questions which are marked “*” is mandatory.
- 4) Finally click on check button to check for any error message.

ENTER THE PRICE OF THE ITEMS IN THE RFx RESPONSE

Enter the Price for all the line items and click on Check if in case of errors and click on Submit to complete the RFx Response.

ASK QUESTIONS TO TVNL IF YOU HAVE ANY DOUBTS REGARDING THE RFx

As a Bidder if You have some questions related to RFx and bidder can also ask questions before submitting the bid.

Step-1: Click on “Questions and Answers (0)” button.

Step-2: Enter the Question and click on Send button to send the question to purchaser/TVNL.

Vendor enters the Message and clicks on “Send” button.

TO UPLOAD THE DOCUMENTS IN THE RFX RESPONSE AS A BIDDER

Step-1: Click on “Notes and Attachments” tab

Step-2: Click on “Collaboration Room” as shown in the below screen to upload the documents.

Step-3: Click on Bidder Document folder.

Step-4: Click on Create Button.

Step-5: Select the Documents to upload the documents

Step-6: Enter the Name, Description of the document and click on Continue

Step-7: Click on Browser button to select the document in your local system.

Step-8: Once the document is selected click on save button.

Step-9: The saved document will be shown.

Step-10: Click on Logout and close the window once you have uploaded all the documents.

TO SUBMIT THE RFx RESPONSE AS A BIDDER

Once you have completed the Price, uploaded the documents click on check button for any errors.

If there is no errors system will show the below message.

Click on Submit button to complete the bidding.

After submitted the system will show the RFx Response status as “Submitted”.

You can change the RFx Response even after bid has been submitted only when the submission date or time is still remaining (Not passed).

Step-1: Open the RFx Response

Step-2: Click on Edit Button

Step-3: Make changes in the RFx Response and Click on Submit.

You can also click in the Withdraw button to withdraw a bid and then If you Click on Edit button after withdrawal you can again make changes and resubmit the RFX Response if the submission time is remaining.

(Signature & Seal of Bidder)



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JUPMI Building premises in ABD Area, Ranchi Smart City, P.O&P.S. Dhurwa, Ranchi-04, (Jharkhand)

GENERAL TERMS & CONDITIONS

- 1. Validity of Offers:** The bid submitted by the bidders should remain valid for a minimum period of 120 days from the date of opening of Part-I unless otherwise specified.
- 2. Prices :-** The Bidder shall quote their rate keeping in view that the rate will remain firm for the entire contract period of 02(two) years.
- 3. Late Receipt of the Tender:** Special attention of the bidder is invited about the receipt of the tender in time. Tenders received after the stipulated date and time due to any reason, whatsoever, shall not be considered. Bidders are requested to see that their bids are delivered in the office of TVNL before the prescribed date and time.
- 4. Inspection of Site:** Before quoting the rate, bidders are advised to visit the work site and get satisfied with scope of work for which rates are to be quoted. Engineer -in -charge may be consulted for any clarification regarding scope of work and terms of tender. Bidder should go through the terms and conditions and technical specifications of work carefully and sign in all pages of the tender document.
- 5. Instruction to Bidder:**
 - a. Bidders are advised to submit their bids strictly based on the scope of work, terms and conditions contained in the bid document and subsequent revisions/amendments, if any. The bid should be submitted along with covering letter.
 - b. The bid shall be prepared and submitted by typing or printing in English on white paper in consecutively numbered pages duly signed by the authorized signatory with company seal affixed on each page. Any part of the bid, which is not specifically signed by the authorized signatory and not affixed with company seal, may not be considered for the purpose of evaluation.
 - c. Copy of bid document, amendments/revisions to bid document if any, shall be signed and submitted along with the bid.
 - d. Complete bid document including all enclosures should preferably be submitted in hard bond or spiral binding and all pages should be numbered and must be signed by the authorized signatory with seal of the company.
 - e. Incomplete tender or tenders not submitted as per requirement as indicated in the NIT are likely to be rejected.
 - f. Bidders shall submit their offer strictly as per terms and conditions of the tender document without any deviation. If at any time any of the documents/information submitted by the bidder is found to be incorrect, false or untruthful, the bid and/or the resultant order may be summarily rejected/ cancelled at the risk of the bidder.
 - g. Failure to furnish all information and documentary evidence as stipulated in the bid document or submission of an offer that is not substantially responsive to the bid document in all respects shall be summarily rejected.
 - h. All bids will be received in duly sealed cover within the due date and time. Bids received after the due date and time is liable for outright rejection.
 - i. Variation may occur in the quantity of items required to be executed under this contract depending on the condition of machine and availability of units.
- 6. Earnest Money Deposit (EMD):** Offers shall be accompanied with Earnest Money Deposit (EMD) of requisite value as specified in a separate envelope failing which offers are liable for rejection. EMD may be furnished in the form of Demand Draft/ Bank Guarantee from any Indian Nationalized Bank/Schedule Commercial Bank in favour of "TENUGHAT VIDYUT NIGAM LIMITED" payable at Ranchi. The Bank Guarantee should remain valid for 12 months. The earnest money will be forfeited on revocation of tender before expiry of validity of the tender or on refusal to enter into the contract agreement after the award is made by TVNL to the bidder.

Tenders without Earnest Money shall not be considered other than NSIC/SSI (Jharkhand) registered units. A copy of such registration has to be furnished for claiming exemption from EMD. Exemption from the Earnest Money to NSIC/SSI units will be considered as per Govt Circulation. SSI units of Jharkhand will be given preference as per Govt rule.

- 7. Security Deposit:** The successful bidder shall have to deposit total security deposit @ **10%** of order value in which initial security money @**5%** of the order value (including earnest money) will be retained as security deposit. Initial security money will be accepted as Bank Draft/ Bank Guarantee in the prescribed format not adjustable against running bill. The security deposit shall have to deposit well before execution of the Agreement. Balance **5%** of security money will be deducted from R/A bills. The security money will be released after successful completion of work and closure of contract.
- 8. Payment Terms:** Payment of Contractor on actual measurement basis will be made through running bills which will be verified by Engineer -in-Charge after successful completion of the work and payment of wages to the employee by the Contractor. The bill in triplicate will be submitted by the Contractor to the Engineer -In-Charge along with Bank Statement and Payment sheet showing payment of wages to workmen and other relevant documents like EPF & GST deposit slip etc before release of payment.
- 9. Paying Authority:** The Accounts Officer, TTPS, Lalpasia will be the Paying Authority.
- 10. Engineer-in-Charge:** Concerned Electrical Executive Engineer will be the Engineer-in-Charge for this work. The works are to be executed as per direction of Engineer-in-Charge. Anything not mentioned here in shall be decided by the Engineer- In- charge of the work and shall be binding on the contractor.
- 11. Period of Contract:** The period of contract shall be 02 (two) years from the date as shall be mentioned in the work order. If the services rendered have been found satisfactory the contract may be extended with mutual consent.
- 12. Safety Measures:** The Contractor shall observe all the safety measures. Safety appliances have to be arranged by the Contractor at his own cost. Any compensation payable to the Contractor's employee in the event of accident or causality has to borne by the Contractor himself and TVNL will not be responsible on this accounts in any way. In case of any damage due to negligence mishandling and mal operation by the staff of contractor, the damage would be made good by the contractor at his own cost. In case of non compliance, proportionate deduction will be made from the bills of the contractor. The contractor shall provide all safety appliances, tools and equipments to his staff and ensure that those items will use by all staff while working and on whole duty period. Name of such important items are given below:-
 - i. Asbestos hand gloves, Apron Asbestos cloths etc.
 - ii. Rubber & Leather hand gloves,
 - iii. Safety shoes for all types of works,
 - iv. Safety belts and hooks,
 - v. Helmets, Ear muffs, Safety goggles, Nose muffs etc.
 - vi. Any other items related to safety rules may be assumed in contractor's scope.
- 13. Scope of the work:** Scope of the work is attached here as a guideline to execute the work. However contractor has to execute all the work as per direction of Engineer-in-Charge for proper working of equipments.
- 14. Labour Laws:** The bidder has to follow the guidelines laid under the labour law, workmen compensation, and EPF rules in respect of minimum wages, EPF payment and other service conditions. In the event of any accident / casualty to the employees of contractor, the compensation payable to the employees is to be borne by the contractor and TVNL will not be responsible in any way.
- 15. Minimum Wage Guarantee:** The contractor shall strictly ensure that the minimum wages as stipulated by GoJ from time to time is paid to their workmen. Any increase in minimum wages during the period of contract, will have to be paid to their workmen without any claim on

TTPS/TVNL. If the contractor fails to comply with guarantee of minimum wages then the agreement is liable to be terminated and security money may be forfeited.

16. Taxes: All statutory taxes as applicable shall be deducted from each running bill of contractor.

17. Violation of Terms & Conditions of Tender: All terms and conditions of tender shall form a part of the contract. If contractor fails to execute the work up to entire satisfaction of Engineer -in -Charge as per terms & conditions of tender, then the contract agreement can be terminated on one month prior notice and security money shall be forfeited.

18. Subletting: The successful bidder will have to execute the work directly. No subletting will be allowed.

19. Competent Authority: During the execution of work in case of any dispute between the Engineer-in -Charge and contractor, the decision of General Manager cum Chief Engineer Tenughat TPS will be final and binding.

20. Man Power: A list of manpower arranged by the contractor should be submitted to Engineer -in -Charge of the work before deployment and same should be deployed with due permission of Engineer -in -Charge. If the performance of the deployed person is not found up to the mark at any time, the contractor will have to replace them with experience and suitable person. During the period of absence of contractors personnel TVNL will have right to deduct penalty from contractor's bill.

21. Tools & Tackles: The contractor should have to arrange minimum of minor and special tools & tackles as per scope of work whereas heavy tools like hydra crane will be arranged by TTPS if available as per request of contractor. Contractor will arrange standard make tools after getting approval from EIC. Contractor will arrange welding M/C, gas cutting set etc to carry out any mechanical work in maintenance of field instruments as and when required. The entry of tools and tackles inside the power house should be verified by the CISF and should match with the requirement. Pro-rata deduction will be made if any mismatch is found.

22. Consumables: The contractor has to provide all consumables like gas, electrodes, cotton waste, markin cloth, rustolin, K-Oil, Petrol/Diesel and abrasive materials at their own cost, if required and applicable. The entry of consumables inside the power house should be verified by the CISF and should match with the requirement. Pro-rata deduction will be made if any mismatch is found.

23. Site Cleaning: The Contractor must keep clean the working area after completion of the job. Waste materials and garbage shall be disposed off by the contractor at suitable place as per the instruction of the Engineer-in-Charge. Old unserviceable spare parts if so, has to be returned by the contractor to the Central Store/Site Store of TTPS as per the instruction of the Engineer-in-Charge.

24. Facilities to be provided by TVNL:

- a. Unfurnished accommodation for office with water and electricity inside the power house will be provided free of cost if available.
- b. The contractor shall be allowed to avail the facilities to extent available at our Chemical Lab at TTPS.
- c. Guest house facility shall be provided to the visiting engineers/officers of the Contractor on chargeable basis if available.

- d. Unfurnished accommodation may be provided to the employees of the Contractor, if available during the period of execution of work on rental basis as per departmental rule.
- e. Electricity will be provided on chargeable basis as per JBVNL Tariff.
- f. Medical facility as available at project site shall also be extended to the employee of the contractor. No payment will be made for any medicine purchased from outside or any treatment or test done outside the TTPS Hospital.

- 25. Working Administration:** In general working hours of the TTPS employee will be the working hours of employee of Contractor. However in case of emergency or as per requirement of the job, contractor's employees will be required to work beyond working hours and also on rest days and holidays on the direction of Engineer -in -Charge. Employees deployed by the contractor will be under the operational control of Engineer -in -charge. In case of any employee working in shift/general duty falling sick or proceeding on leave the facts shall be brought to the knowledge of Engineer -in -Charge and deficit will be made by the contractor either by the existing staff by performing overtime duty or getting other manpower without adding any additional cost to the Project. If the work gets hampered due to deficit in manpower, the penalty will be deducted from the contractor's bill. It shall be binding on contractor that it does not directly or through its personnel divulge the secrecy of the establishment of TTPS. Any person deployed by contractor will not be a member of any trade union operating in the TTPS Projects establishment or outside. They will not take part in any strike or sympathized with any such movement.
- 26. Penalty:** In case of any breakdown, fault or emergency maintenance, the contractor has to attend the same without any delay. In case of failing or deliberated neglect, the work will be carried out through outside agency at the cost & risk of the contractor and this may be liable for termination of their contract. A poor (Manpower: skilled or unskilled, T&P and consumables) mobilization during OH or Emergency jobs shall attract a penalty of Rs 5000/ per day. Further any safety lapses like non use of PPEs, improper handling of gas cylinders, improper power connection, damaged electrical equipments, non use of plug tops, inadequate scaffolding where work is being carried out at height, poor/absence of temporary fencing where railing has been cut to facilitate lifting up or/and down of materials etc shall attract a penalty of Rs 5000/ per occasion. Scrap generated during the work shall be removed every day(as directed by EIC) by the contractor to keep the area clean. If scrap is not removed then the same will be removed at risk and cost of agency @ Rs 5000/MT will be deducted as penalty.
- 27. Departmental Facilities:** In case of major repair of equipments and testing of special equipments contractor should have to provide services of an expert from manufacturer and special tools & tackles. The contractor will also provide all other technical staffs and T&P's consumables etc free of cost. However, all spares parts shall be provided by TVNL on free of cost.
- 28. Transportation:** Successful bidder will be required to arrange his own conveyance for their employee for attending their duties.
- 29. Mobilization Time:** The successful bidder will be given maximum 15 days time from the date of LOI / Work Order for Mobilization of resources, manpower to start the work. However for the work under ARC, contractor shall be given the mobilization period of one day. The contractor should be in position to ensure mobilization even within shorter period to start the work if need so arises like availability of the shutdown of units. No extra claim shall be entertained by either for idle labour or reduced mobilization period.
- 30. Agreement:** The firm or his legally authorized representative having Power of Attorney will have to execute an agreement with the ordering authority within 10 days of issue of order in the event of placement of order on them. The agreement will have to be prepared on a non-judicial stamp paper of Rs.50/- only on Nigam's format. Bank Guarantee against Security Deposit, Performance Guarantee etc. as per order will have to be furnished well before execution of agreement. The contractor is required to produce all original paper and documents for verification before agreement failing which order will be cancelled. Contractor will not be allowed to start the work without executing the agreement.
- 31. Jurisdiction of Court:** Any dispute between the Contractor and TVNL shall be settled in the Civil Court, Ranchi, Jharkhand only.

(Signature & Seal of Bidder)



TENUGHAT VIDYUT NIGAM LIMITED

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JUPMI Building premises in ABD Area, Ranchi Smart City, P.O&P.S. Dhurwa, Ranchi-04, (Jharkhand)

NIT No. 30/BMD/W/TVNL/RAN/2026-27, SRM RFx No-1-14024

Sub: 1. Annual Rate Contract for Boiler Pressure Parts & H.P. Valves of Boiler of Unit No. 1 & 2 at TTPS, Lalpania for the period of two years. 2. Annual Rate Contract for APH, Scanner Fan, Dampers and Gates of Boiler of Unit No. 1 & 2 at TTPS, Lalpania for the period of two years.

Scope of work

ANNUAL RATE CONTRACT FOR BOILER PRESSURE PARTS & H.P. VALVES OF BOILER OF UNIT NO-1 & 2 FOR THE PERIOD OF TWO YEARS

ITEM-1: MOBILIZATION OF MAN-POWER FOR ATTENDING TUBE LEAKAGE OR DURING SHORT SHUTDOWN:

- A.) OPENING OF MAN-HOLES AS REQUIRED BY EIC.
- B.) SHIFTING OF SCAFFOLDING MATERIALS, ALL T&P CUTTING & WELDING ACCESSORIES ETC. TO THE WORK PLACE.
- C.) ASSISTANCE IN DETECTION OF TUBE LEAKAGE/DEFECTS.
- D.) FOR ATTENDING TUBE LEAKAGE/DEFECTS ANY CUTTING/REMOVAL OF PLATE, REFRACTORY, LIFTING OF ANY COIL BANK ETC. IS REQUIRED TO BE DONE BY CONTRACTOR AND SAME TO BE RESTORED /LOWERED AFTER ATTENDING THE JOB.
- E.) SHIFTING OF TUBES, BENDS, SHIELDS, LOCKS, CLAMPS, ETC.
- F.) SUFFICIENT & COMPETENT MAN-POWER WITH ALL TOOL & TACKLES FOR ATTENDING TUBE LEAKAGES IS TO BE MOBILIZED WITHIN 6 HOURS OF TELEPHONIC INFORMATION BY EIC.
- G.) FAILING TO ABOVE, SUITABLE PENALTY WILL BE IMPOSED BY TVNL
- H.) WORK DURING SHORT SHUTDOWN CARRIED OUT AS PER THE DIRECTION OF EIC, IF CONTRACTOR DENY/ FAILS TO DO SO THEN PENALTY WILL BE IMPOSED EQUAL TO 05(FIVE) TIMES OF RATE OF THE SPECIFIED JOB.

ITEM -2: ERECTION OF SKY CLIMBER AS PER REQUIREMENT & ITS REMOVAL AFTER WORK.

- A.) SHIFTING OF SCAFFOLDING MATERIAL & SKY CLIMBER LIKE PLANKS, PIPES, SKY CLIMBER MATERIAL FROM STORES TO WORK SITE AND TRANSPORT BACK TO STORES AFTER DISMANTLING.
- B.) OPENING OF FURNACE MAN HOLES FOR ENTERING THE FURNACE.
- C.) ERECTION OF PLATFORM INSIDE THE FURNACE BOTTOM FOR SKY CLIMBER INSTALLATION. SAFETY OF THE EQUIPMENTS AND THE PERSONAL IS VERY IMPORTANT & HENCE ERECTION TO BE DONE UNDER STRICT TRAINED SUPERVISOR

D.) ASSEMBLY AND ERECTION OF SKY CLIMBERS IN ANY WALL / PLATEN AS PER REQUIREMENT GIVEN BY EIC. THE OPERATOR SHOULD BE SKILLED IN MOVING THE SKY CLIMBER AND KNOWS THE SAFETY OPERATIONS ALSO.

E.) DISMANTLING OF SKY CLIMBERS, PLATFORM AFTER COMPLETION OF WORK.

F.) INSPECTION, CHECKING AND MECHANICAL MAINTENANCE OF SKY CLIMBER EQUIPMENT AFTER EACH INSTALLATION.

G.) SKY CLIMBER SHALL BE KEPT CLEAN DURING USE BY AGENCY.

H.) AGENCY SHALL ENGAGE QUALIFIED AND EXPERIENCED OPERATOR FOR SKY CLIMBER REQUIREMENT. MIN 2 PERSON SHALL BE THERE WHILE SKY CLIMBER OPERATION IS ON.

I.) OPERATION OF SKY CLIMBER FOR FURNACE CLEANING, INSPECTION, REPAIR, REFRACTORY, BURNER RELATED WORKS, WALL BLOWER ADJUSTMENT OR ANY OTHER WORK.

J.) AGENCY SHALL COMPLETE AND MAKE SKY CLIMBER OPERATIONAL WITHIN 04 HRS OF ISSUE OF PTW IN CASE OF UNIT S/D OTHER THAN O/H WORKS.

K.) ANY DELAY OR NON AVAILABILITY OF TRAINED PERSONNEL WILL CALL FOR SUITABLE PENALTY.

L.) ALL CONSUMABLES LIKE ROPE, HAND LAMPS, AND ELECTRODE SHOULD BE PROVIDED BY THE CONTRACTOR AT ITS OWN COST.

M.) ALL SORTS OF TRANSPORT AND MEANS OF TRANSPORT IS IN SCOPRE OF CONTRACTOR.

N.) SCRAP MATERIAL GENERATED DURING WORK SHALL HAVE TO BE TRANSPORTED TO SCRAP YARD BY THE CONTRACTOR.

O.) LOAD TESTING OF SKY CLIMBER WILL HAVE TO BE DONE BY CONTRACTOR ONCE IN A YEAR.

ITEM-3: ROTATION OF SKY CLIMBER TO OTHER SIDE OF W/W.

A.) CHANGING OF SKY CLIMBER POSITION IN OTHER WALL (I.E. SHIFTING OF PLATFORMS AND POSITIONS OF ROPES AND END PLATFORM) AS PER ENGINEER-IN-CHARGE.

B.) TIME FOR CHANGING- 03 HRS.

C.) NO COMPROMISE TO BE MADE IN SAFETY OF THE EQUIPMENT AND CHANGING/OPERATION OF THE SKY CLIMBER TO BE DONE UNDER STRICT SUPERVISION OF TRAINED AND SKILLED PERSONNEL.

ITEM - 4: ERECTION OF TEMPORARY SCAFFOLDINGS INSIDE FURNACE FOR SKY CLIMBER AND IN RH/PSH/ECONOMIZER HOPPER ZONE FOR REACHING THE WORK AREA.

- A.) TRANSPORTATION OF MATERIAL LIKE PLANKS, SCAFFOLDING MATERIAL FROM STORE TO SITE.
- B.) IF SCAFFOLDING IS TO BE ERECTED INSIDE BOILER, OPENING OF ELLIPTICAL MAN-HOLES & OTHER MAN-HOLES REQUIRED FOR FIXING OF SCAFFOLDING.
- C.) ALL CONSUMABLES LIKE ROPE, HAND LAMPS, AND ELECTRODE SHOULD BE PROVIDED BY THE CONTRACTOR AT ITS OWN COST.
- D.) ON GETTING INTIMATION CONTRACTOR SHOULD MOBILISE ITS MAN-POWER WITHIN THREE HOURS.
- E.) MAN-POWER DEPLOYED IN SCAFFOLDING MAY BE UTILISED SOMEWHERE ELSE TO TAKE UP THE RELATED JOBS E.G. MATERIAL TRANSPORTATION OR ASSISTANCE TO OTHER JOBS.
- F.) ALL SORTS OF TRANSPORT AND MEANS OF TRANSPORT IS IN SCOPRE OF CONTRACTOR.
- G.) THE WORK PLACE MUST BE KEPT CLEAN.
- H.) THE PARTY SHALL ARRANGE NECESSARY AREA LIGHTINGS REQUIRED FOR CARRYING OUT JOB. ONLY FLOOD LIGHT TO BE PROVIDED BY TVNL.
- I.) REMOVE SCAFFOLDINGS AFTER COMPLETION OF WORK.

ITEM-5: ERECTION OF TEMPORARY SCAFFOLDINGS (SMALL) FOR REACHING THE WORK AREA.

- A.) ERECT PROPER SCAFFOLDINGS & PLATFORM FOR REACHING THE WORK AREA EXCEPT IN RH/PSH ZONE FOR ATTENDING TUBE LEAKAGE IN A SAFE MANNER.
- B.) TRANSPORTATION OF MATERIAL LIKE PLANKS, SCAFFOLDING MATERIAL FROM STORE TO SITE.
- C.) IF SCAFFOLDING IS TO BE ERECTED INSIDE BOILER, OPENING OF ELLIPTICAL MAN-HOLES & OTHER MAN-HOLES REQUIRED FOR FIXING OF SCAFFOLDING.
- D.) ALL CONSUMABLES LIKE ROPE, HAND LAMPS, AND ELECTRODE SHOULD BE PROVIDED BY THE CONTRACTOR AT ITS OWN COST.
- E.) ON GETTING INTIMATION CONTRACTOR SHOULD MOBILISE ITS MAN-POWER WITHIN THREE HOURS.
- F.) MAN-POWER DEPLOYED IN SCAFFOLDING MAY BE UTILISED SOMEWHERE ELSE TO TAKE UP THE RELATED JOBS E.G. MATERIAL TRANSPORTATION OR ASSISTANCE TO OTHER JOBS.

G.) ALL SORTS OF TRANSPORT AND MEANS OF TRANSPORT IS IN SCOPRE OF CONTRACTOR.

H.) THE WORK PLACE MUST BE KEPT CLEAN.

I.) THE PARTY SHALL ARRANGE NECESSARY AREA LIGHTINGS REQUIRED FOR CARRYING OUT JOB. ONLY FLOOD LIGHT TO BE PROVIDED BY TVNL.

J.) REMOVE SCAFFOLDINGS AFTER COMPLETION OF WORK.

ITEM - 6: REPLACEMENT OF TUBES AND VALVES (Tube O.D.= Ø 44.5 mm to Ø 76.1 mm)

FOR REPLACEMENT OF TUBE:

A.) ERECTION OF SCAFFOLDING AS PER REQUIREMENT.

B.) CUTTING OF DEFECTIVE TUBES MARKED FROM LOCATION AS PER IBR NORMS. CUT EDGES OF THE TUBES ARE TO BE PROPERLY COVERED SO THAT NO FOREIGN MATERIAL CAN GO INSIDE THE TUBE. COILS MAY HAVE TO BE LOWERED/PULLED SIDEWAYS/LIFTED FROM ITS POSITION, ETC. FOR ACCESSING THE TUBES TO BE CUT.

C.) CUTTING AND EDGE PREPARATION OF SPOOL PIECES ON LENGTHS OF DAMAGED TUBE BY PROCEDURE PRESCRIBED AS PER WELDING CODE.

D.) EDGE PREPARATION OF JOINTS AS PER IBR NORMS.

E.) FITTING OF TUBES AND OFFER THE SAME FOR INSPECTION.

F.) WELDING OF JOINTS BY ROOT TIG AND SUBSEQUENT ARC WELDING BY PROCEDURE PRESCRIBES AS PER WELDING CODE. FOR THIS WORK BUTT JOINT OF TUBE, FILLET CUM GROOVE JOINT OF HEADER TO TUBE ETC. MAY HAVE TO BE DONE.

G.) REFITTING ALL THE ATTACHMENTS /FINS AS REQUIRED.

H.) ELECTRODES BAKING OVENS ARE TO BE USED.

I.) ANY PRESSURE PARTS WELDING COMING UP IN THE NON- SCHEDULE PIPING WILL BE PAID ON PROPORTIONAL BASIS OF WATER WALL TUBES.

J.) ARRANGE CUTTING SET, GRINDING MACHINE, ARGON SET AND WELDING MACHINE ETC.

K.) SHIFT THE REQUIRED MATERIALS TO THE REQUIRED LOCATION OF THE BOILER FOR ATTENDING THE JOB.

L.) CUTTING OF THE MARKED/ DAMAGED TUBES BY CUTTING SET.

M.) PLACEMENT OF NEW TUBE PIECES AFTER GRINDING & MAKING 'V' WITH PROPER ALIGNMENT. JOINING OF TUBES BY H.P. WELDING. JOINT WELDING FULL TIG /ROOT TIG WILL BE CARRIED AS PER THE REQUIREMENT.

- N.) WELDING ELECTRODES OF REPUTED MAKE LIKE ADOR/L & T/ D & H/ ESAB SHOULD BE ARRANGED BY CONTRACTOR.
- O.) FABRICATE REQUIRED SIZE BENDS FROM STRAIGHT TUBES, IF READY-MADE BENDS ARE NOT AVAILABLE.
- P.) NO EXTRA PAYMENT SHALL BE MADE FOR FIN CUTTING OF THE DAMAGED TUBES.
- Q.) FABRICATE REQUIRED SIZE FINS FROM M.S. / C.S. PLATE, IF READY-MADE FINS ARE NOT AVAILABLE.
- R.) THE HEALTHINESS OF FLEXIBLE CONNECTORS FOR ALLOWING AXIAL MOVEMENT AND MAINTAINING TUBE ALIGNMENT SHALL BE THOROUGHLY INSPECTED.
- S.) ANY DAMAGE FLEXIBLE CONNECTORS MUST BE REPLACED OR CORRECTED FOR FREE MOVEMENT OF THE ASSEMBLY.
- T.) PAYMENT FOR FIN WELDING SHALL BE MADE ON THE BASIS OF RM AS PER ITEM NO.-7.
- U.) TUBE REPAIR/REPLACEMENT (UP TO 76.1 MM OD).
- FOR REPLACEMENT OF VALVES:
- A.) PROVIDE SUPPORT TO THE PIPELINE
- B.) AFTER REMOVAL OF INSULATION AND ERECTION OF SCAFFOLDING REMOVE THE ACTUATOR. CUT THE DEFECTIVE VALVE.
- C.) PREPARE THE EDGE OF THE CONNECTING PIPELINES AS PER EIC
- ITEM-7: SKIN CASING WELDING IN PENT HOUSE & OTHER AREA OF BOILER
- A.) REMOVAL/ REPLACEMENT/ SET-RIGHTING THE DAMAGED/ BOWED M.S. CHANNELS/ BEAMS.
- B.) PLACEMENT OF 3.15 MM M. S. PLATES/C.S. PLATES IN PLACE OF DAMAGED ONES.
- C.) REMOVAL OF DEBRIS & CLEANING THE SITE.
- D.) WELDING OF BOILER TUBE FINS, REPAIRING OF WIND BOX, THEIR EXPANSION JOINTS.
- E.) THE INSPECTION OF WIND BOX SUPPORTS AND ITS EXPANSION JOINTS, BURNER BLOCK CONNECTING LINKS, S-PANEL SUPPORTS AND THEIR RECTIFICATIONS/ REPLACEMENTS OF ALL ABOVE AS REQUIRED.
- F.) DURING THE CHECKING OF THE SEISMIC GUIDE AND THE FURNACE GUIDES, THE GAPS ARE TO BE MEASURED AND RECORDED.
- G.) PAYMENT FOR WELDING SHALL BE MADE ON THE BASIS OF RM.

- H.) FABRICATE REQUIRED SIZE FINS FROM M.S./ C.S. PLATE, IF READY-MADE FINS ARE NOT AVAILABLE.
- I.) THE HEALTHINESS OF FLEXIBLE CONNECTORS FOR ALLOWING AXIAL MOVEMENT AND MAINTAINING TUBE ALIGNMENT SHALL BE THOROUGHLY INSPECTED.
- J.) ANY DAMAGE FLEXIBLE CONNECTORS MUST BE REPLACED OR CORRECTED FOR FREE MOVEMENT OF THE ASSEMBLY.
- K.) PAYMENT FOR FIN WELDING SHALL BE MADE ON THE BASIS OF RM.
- L.) FOR REPLACEMENT OF CROWN PLATES, SEAL BENDS ETC,IF ANY, EXTRA WORK LIKE LIFTING AND LOCKING OF COILS REQUIRED,IT IS TO BE DONE BY CONTRACTING AGENCY.
- M.) INSPECTION DURING AIR TIGHTNESS TEST AND RECTIFY THE DEFECT IF ANY.
- N.) ELECTRODE TO BE DRIED AS PER NORMS FOR WELDING OF PLATES. ELECTRODE-7018 TO BE USED FOR WELDING OF SKIN CASING WELDING. PENALTY WILL BE IMPOSED IN USE OF ANY LOW QUALITY OF ELECTRODE.

ITEM-8: CASSETTE BAFFLE REPLACEMENT/SETTING (AFTER REPAIRING)

- A.) ENSURE PTW.
- B.) DISMANTLING & REMOVAL OF DEFORMED CASSETTE BAFFLES.
- C.) RECTIFICATION OF DEFECTS OF DEFORMED CASSETTE BAFFLES BY HEATING/ PULLING/HAMMERING ETC.
- D.) REFITTING OF CORRECTED CASSETTE BAFFLES.
- E.) PLACEMENT OF NEW CASSETTE BAFFLES (IF AVAILABLE).
- F.) FOUR NOS. OF CASSETTE BAFFLE ALIGNMENT IS EQUAL TO REPAIRING/REPLACEMENT OF ONE CASSETTE BAFFLE.

ITEM-9: APPLICATION OF PROTECTIVE SHIELD ON TUBES.

- A.) ENSURE PTW.
- B.) FIX S.S. SHIELDS OVER TUBES WITH PROPER WELDING AS & WHEN REQUIRED AS PER THE INSTRUCTION OF E-L-C.
- C.) REMOVE OLD DAMAGED SHIELDS AND FIX NEW SHIELDS/RODS.
- D.) ERECT SCAFFOLDING FOR SHIELD FIXING WORK AS PER REQUIREMENT.
- E.) THE SHIELD/ROD WILL BE PROVIDED BY TTPS FREE OF COST IN LENGTH OF 500MM TO 1000MM.

ITEM-10: MANIPULATION OF BENDS (SHORT/LONG, OFFSETS).

- A.) MANUFACTURE OF BEND AS PER DRAWING/SAMPLE.
- B.) INSPECTION OF BEND BY TVNL REPRESENTATIVE AS PER LAID DOWN PROCEDURE AND QUALITY NORMS.
- C.) BENDS TO BE FABRICATED FROM THE TUBE SIZE OF 38 MM O.D. TO 76.1.
- D.) FOR MANUFACTURE OF BEND, TUBE BENDING MACHINE, DYE AND ALL OTHER REQUIRED ATTACHMENTS ARE IN THE SCOPE OF CONTRACTOR.

ITEM-11: REPLACEMENT OF GASKETS OF BOTH DOORS OF BOILER DRUM

- A.) OPENING OF MAN-HOLE DOOR OF BOILER DRUM.
- B.) REMOVAL OF OLD/ DAMAGED GASKETS.
- C.) CUTTING NEW GASKET (THICK: 3MM, METALLIC), AS PER SIZE.
- D.) PLACEMENT OF GASKETS AT BOTH DOORS OF BOILER DRUM.
- E.) APPLICATION OF SUITABLE ADHESIVE.
- F.) GASKET SHEET WILL BE SUPPLIED BY TTPS FREE OF COST.
- G.) CLOSING OF MAN-HOLE DOOR OF BOILER DRUM.
- H.) TIGHTENING OF BOLTS.
- I.) HOT- TIGHTENING OF BOLTS.

ITEM-12: REFRACTORY WORK.

- A.) DAMAGED REFRACTORY TO BE BROKEN AS DECIDED BY EIC.
- B.) ALL DAMAGED PLATES BELOW REFRACTORY TO BE REPLACED BY NEW ONE.
- C.) CASTING OF REFRACTORY IF REQUIRED TO BE DONE BY AGENCY.
- D.) WATER SPRAY TO BE DONE AFTER CERTAIN INTERVAL FOR CURING.
- E.) TRANSPORTATION OF REFRACTORY FROM STORE TO SITE IS IN THE SCOPE OF CONTRACTOR.

ITEM-13: SERVICING OF BOILER ELLIPTICAL MANHOLE, MANHOLE, AND PEEPHOLE DOOR:

- A.) OPEN ALL MANHOLES, PEEP HOLES.
- B.) INSPECTION OF ALL MANHOLES, PEEP HOLES.
- C.) SERVICING OF ALL MANHOLES, PEEP HOLES.

- D.) BROKEN/MISSING HINGES, HINGES PINS, SCREWS, PINS, HAND WHEELS TO BE REPLACED/ SERVICED AS DIRECTED BY EIC.
- E.) CHECKING OF FREENESS OF ALL THE PEEP HOLES, MANHOLE.
- F.) REFRACTORY TO BE CASTED ON INSIDE LAYERS OF MANHOLES, PEEP HOLES.
- G.) REPLACEMENT OF ROPES ON MANHOLE TO PREVENT LEAKAGE OF FLUE GAS.
- H.) INSPECTION DURING AIR TIGHTNESS TEST AND RECTIFY THE DEFECT IF ANY.
- I.) IF REQUIRED REFRACTORY TO BE CASTED ON INSIDE LAYERS OF MANHOLES, PEEP HOLES.
- J.) REPAIRING/REPLACEMENT OF EXPANSION RODS OF BOILER.

ITEM 14: PAD WELDING ON STUBS.

- A.) MAKE PROPER APPROACH TO TAKE UP THE JOB.
- B.) INSPECT THE STUBS OF THE VARIOUS SUPER HEATER HEADERS.
- C.) ENGAGE IBR CLASS WELDER (CERTIFIED IBR WELDER WITH TIG WELDING EXPERIENCE AND HAVING VALID LICENSE FOR WELDING OF CARBON STEEL AND ALLOY STEEL FROM CHIEF INSPECTOR OF FACTORIES AND BOILER JHARKHAND) FOR FILLING UP OF THE ERODED STUBS OF SUPER HEATER HEADERS BY H.P. WELDING.
- D.) MOBILIZE NECESSARY MANPOWER AND MATERIALS FOR PAD WELDING OF STUBS OF SUPER HEATER HEADER.
- E.) ARRANGE GRINDING MACHINE, ARGON SET, AND OVEN FOR HEATING WELDING RODS AND WELDING MACHINE ETC.
- F.) FILLING UP OF THE STUBS BY H.P.WELDING.
- G.) WELDING ELECTRODES OF REPUTED MAKE LIKE ADOR/L & T/ D & H/ ESAB SHOULD BE ARRANGED BY CONTRACTOR.
- H.) NO EXTRA PAYMENT SHALL BE MADE FOR FIN CUTTING NEAR ERODED STUBS ON WHICH PAD WELDING IS TO BE DONE.

ITEM 15: CLEANING OF 1ST PASS OF BOILER BY FIRE WATER:

- A.) 1ST PASS OF BOILER WALL BLOWER, BURNER AREA, PLATEN & REHEATER COILS TO BE THOROUGHLY WASHED AND CLEANED.
- B.) WASHING TO BE DONE BY WATER HOSE AND NOZZLE IS IN THE SCOPE OF CONTRACTOR.

C.) FOR THIS JOB, MANPOWER DEPLOYED ROUND THE CLOCK.
CONTRACTOR SHOULD DEPLOY HIS MEN IN SHIFT.

D.) THE PARTY SHALL ARRANGE NECESSARY AREA LIGHTING REQD FOR
CARRYING OUT JOB. ONLY FLOOD LIGHT TO BE PROVIDED BY TVNL FREE OF
COST.

E.) SAFETY PRECAUTION TO BE TAKEN DURING WASHING OF BOILER.

ITEM 16: CLEANING OF 2ND PASS OF BOILER DURING SHORT SHUT DOWN:

A.) SECOND PASS I.E. FINAL SH, LTSH, ECONOMISER AREA TO BE
THOROUGHLY WASHED AND CLEANED.

B.) WASHING TO BE DONE BY WATER (HOSE AND NOZZLE) IS IN THE SCOPE
OF CONTRACTOR.

C.) CONTRACTOR SHOULD DEPLOY MANPOWER ROUND THE CLOCK IN
SHIFT

D.) THE PARTY SHALL ARRANGE NECESSARY AREA LIGHTING REQD. FOR
CARRYING OUT THE JOB. ONLY FLOOD LIGHT WILL BE ARRANGED BY TVNL
FREE OF COST.

E.) SAFETY PRECAUTION TO BE TAKEN DURING WASHING OF BOILER.

ITEM 17: CLEANING OF PENT HOUSE AREA/ DRUM HOUSE OF BOILER DURING
SHORT SHUTDOWN:

A.) PENT HOUSE AREA/ DRUM HOUSE OF BOILER TO BE THOROUGHLY
CLEANED.

B) THE ASH SHOULD BE DISPOSED FROM PENT HOUSE AREA/ DRUM HOUSE
OF BOILER TO ZERO METER OF BOILER THROUGH PROPER ARRANGEMENT,
ASH SHOULD NOT BE SCATTERED IN BOILER AND AROUND THE BOILER AREA
AFTER THAT ASH SHOLUD BE DISPOSED TO ASH DISPOSAL YARD .

C) ALL THE ASH SPREADES AT ZERO METER SHOULD BE COLLECTED AND
DISPOSED TO ASH DISPOSAL YARD.NO EXTRA PAYMENT SHOLUD BE MADE
FOR DISPOSAL OF ASH TO ASH DISPOSAL YARD.

E.) FOR THIS JOB, MANPOWER DEPLOYED ROUND THE CLOCK.
CONTRACTOR SHOULD DEPLOY HIS MEN IN SHIFT.

D.) THE PARTY SHALL ARRANGE NECESSARY AREA LIGHTING REQD FOR
CARRYING OUT JOB. ONLY FLOOD LIGHT TO BE PROVIDED BY TVNL FREE OF
COST.

E.) SAFETY PRECAUTION TO BE TAKEN DURING CLEANING OF PENT HOUSE
AREA/ DRUM HOUSE OF BOILER.

ITEM 18: SERVICING OF FUEL OIL PUMPS:

A.),,REMOVE THE PUMP FROM ITS POSITION.

B.),,COMPLETE DISMANTLING OF PUMP.
 C.),,CLEANING AND INSPECTION OF ALL THE COMPONENTS.
 D.),,REPAIR/REPLACE ANY COMPONENT AS DIRECTED BY EIC
 E.),,ASSEMBLE THE PUMPS.
 F.),,MANUALLY ROTATE THE PUMP AND CHECK FOR ANY ABNORMALITIES RECTIFY THE SAME.
 G.),,FIXING OF PUMP IN POSITION.
 H.),,CHECKING OF HEALTHINESS OF COUPLIN G, REPLACE/REPAIR THE SAME.
 I.),,ALIGNMENT OF PUMP WITH MOTOR.
 J.),,TRIAL RUN OF PUMP.

ITEM 19: REMOVAL OF WORN OUT COAL COMPARTMENT ASSEMBLY, THEIR REPAIRING, REFITTING, INCLUDING SETTING OF BURNER TILT MECHANISM FOR THEIR SYNCHRONOUS OPERATION:-

- A.) REMOVE THE BOILER INLET GATE AFTER REMOVAL OF BOILER INLET ELBOW.
- B.) REMOVE SCANNER ASSEMBLIES, OIL GUN ASSEMBLIES AND IGNITER ASSEMBLIES.
- C.) REMOVE ALL COOLING HOSES.
- D.) REMOVE THE INNER PANEL & OUTER PANEL ASSEMBLIES.
- E.) REMOVE COAL COMPARTMENT ASSEMBLIES.
- F.) REPAIR THE BURNER TIP & TAIL PIECE WITH SUITABLE ELECTRODE.
- G.) REMOVE OIL NOZZLES, END AIR NOZZLES & AUX. AIR NOZZLES.
- H.) CLEAN ALL THE NOZZLE ASSEMBLIES.
- I.) REMOVE DAMAGED OLD FASTENERS AND FIX NEW FASTENERS.
- J.) PUT NEW/RECONDITIONED COAL COMPARTMENT ASSEMBLIES AS PER INSTRUCTION OF E-I-C.
- K.) PUT OIL NOZZLES, END AIR NOZZLES & AUX. AIR NOZZLES.
- L.) PUT NEW INSULATION ON THE OUTER PANELS.
- M.) ASSEMBLE THE INNER & OUTER PANELS.
- N.) PUT NEW CERAMIC TAPE, GASKETS FOR ASSEMBLING.
- O.) ARRANGE ALL TOOLS & TACKLES, CUTTING / WELDING REQUIRED FOR THE ABOVE WORK.
- P.) SHIFT DAMAGED NOZZLES (COAL NOZZLE AND TIP) TO A SUITABLE LOCATION AS PER THE INSTRUCTION OF E-I-C.
- Q.) DISMANTLE THE BURNER TILT MECHANISM IN ALL THE FOUR CORNERS.

R.) CLEAN ALL THE PARTS.

S.) CHECK ALL THE PINS FOR ANY WEAR AND TEAR.

T.) REPLACE ALL THE DAMAGED PINS. APPLY ANTI SEIZE SPRAY ON ALL THE CONTACT AREAS. CHECK ALL THE DRIVE SHAFT TRUENESS. REASSEMBLE THE TILT MECHANISM WITH NEW SPLIT PINS. MANUALLY CHECK THE OPERATION FOR TILT. CONNECT ALL THE REACH RODS TO THE TILT MECHANISM.

U.) CARRYOUT THE ZERO SETTING AFTER NOZZLES IN ALL THE ELEVATION IN ALL THE FOUR CORNERS IS INSERTED. LOCK ALL THE COAL COMPARTMENT ASSEMBLY AFTER ZERO SETTING. REPAIR/REPLACE DAMAGED HOUSINGS SO THAT ALL THE HOUSINGS ARE IN FLEXIBLE CONDITION SO THAT IT WILL NOT AFFECT THE TILT.

V.) REPAIR THE SS PLATE COVERING ON THE BURNER PANEL TUBES BY CUTTING & WELDING SO THAT IT IS WILL NOT FOUL WITH NOZZLE TIP DURING TILTING. CLEAN ALL THE SCANNER HOUSINGS, IGNITER HOUSINGS AND OIL GUN HOUSINGS. CONNECT THE POWER CYLINDER TO THE TILT MECHANISM.

W.) OPERATE POWER CYLINDER TO CHECK THE TILT (± 30 DEGREE).

X.) CARRYING OUT SERVICING OF OIL GUN ASSEMBLY

ITEM 20: ATTENDING OF VALVE GLAND LEAKAGE /PRESSURE SEAL LEAKAGE/ BONNET

LEAKAGE/LINE FLANGE LEAKAGE BY TIGHTENING OR REPLACING GLAND OR GASKET, SERVICING OF VALVE WORKING NUT:

LEAKAGE ARREST BY TIGHTENING GLAND, PRESSURE SEAL RING OR BONNET OF VALVES OR VALVE FLANGE OR LINE FLANGE. IF THERE IS A LEAKAGE FROM VALVE GLAND/PRESSURE SEAL RING/ BONNET OR IN LINE FLANGE, TIGHTENS THE GLAND FOLLOWER/BOLT/BONNET/FLANGE IMMEDIATELY AND ARREST THE LEAKAGE. PAYMENT FOR ONE NO MEANS ARRESTING FIRST ONE TO FIVE NO LEAKAGE IN A DAY. ABOVE FIVE NO, THREE LEAKAGE ARRESTING WILL CONSIDER ONE NO. IF ONE DAY ONLY ONE LEAKAGE ARRESTS IT WILL CONSIDER AS ONE NO.

OR

REPLACE OF GLANDS

A) REMOVE THE NUTS OF GLAND FOLLOWER

B) EXTRACT THE OLD GLAND PACKING RINGS/ROPE FROM THE STUFFING BOX

C) INSERT NEW GLAND PACKING RINGS/ROPE AS PER INSTRUCTION OF EIC

D) TIGHTEN THE GLAND FOLLOWER NUTS.

E) CHECK FREENESS OF VALVE SPINDLE AFTER GLAND FOLLOWER TIGHTENING.

OR

REPLACEMENT OF PRESSURE SEAL RING /BONNET GASKETS

A) AFTER REMOVING ALL CABLE/C&I CONNECTIONS OF THE ACTUATORS DECOUPLE THE ACTUATOR ASSY. AND REMOVE THE SAME FROM POSITION.

B) DISMANTLE THE VALVE COMPLETELY

C) REPLACE THE PRESSURE SEALING RINGS/SEAL RINGS/GASKET

D) THOROUGHLY CLEAN AND SERVICE THE COMPONENTS SERVICE THE WORKING NUT AND BOX UP THE VALVE.

E) ASSEMBLE THE VALVE AND INSERT NEW GLAND PACKINGS.

F) CHECK THE MECHANICAL FREENESS OF THE VALVE AND REMOUNT THE ACTUATOR.

OR

REPLACEMENT OF LINE GASKET

A) LOOSEN THE FLANGE FASTENERS.

B) REMOVE THE OLD GASKETS.

C) INSERT THE NEW GASKET AND TIGHTEN FASTENERS.

OR

SERVICING OF WORKING NUT

A) AFTER REMOVING ALL CABLE/C&I CONNECTIONS OF THE ACTUATORS DECOUPLE THE ACTUATOR ASSY. AND REMOVE THE SAME FROM POSITION.

B) REMOVE THE WORKING NUT AND BEARINGS.

C) CLEAN AND INSPECT THE SAME FOR IDENTIFICATION OF DAMAGE/DEFECT.

D) REPAIR/REPLACE THE DEFECTIVE COMPONENTS.

E) APPLY FRESH LUBRICANT AND REASSEMBLE THE WORKING NUT OF THE VALVE.

ITEM-21: REPLACEMENT OF SAMPLE COOLER

A.) TRANSPORTATION OF NEW/REPAIRED SAMPLE COOLER FROM CENTRAL/ SITE STORE TO WORKING SITE.

B.) CUTTING & REMOVAL OF DEFECTIVE SAMPLE COOLER.

- C.) PREPARING THE SURFACE BY GRINDING & MAKING 'V'/'J'.
- D.) JOINING THE NEW/REPAIRED SAMPLE COOLER WITH THE RESPECTIVE PIPE LINE BY HIGH PRESSURE WELDING.
- E.) TRANSPORTATION OF DEFECTIVE SAMPLE COOLER, AS PER DIRECTION OF EIC.

OR

REPAIRING OF DAMAGED SAMPLE COOLER-

- A.) CLEANING OF THE COOLER. CUTTING OF THE TOP PORTION OF THE SHEEL OF COOLER FOR INSPECTION.
- B.) CHECKING OF THE S.S TUBE COIL INSIDE THE COOLER FOR ANY LEAKAGE.
- C.) REPLACEMENT OF THE DAMAGED PORTION OF THE S.S. TUBE/COMPLETE S.S. TUBE COIL AS PER REQUIREMENT.
- D.) CARRYING OUT HYDRAULIC TEST OF COOLER.
- E.) WELDING OF THE CUT PORTION/ MINOR REPAIRING OF SHEEL PORTION BY H.P. WELDING AS REQUIRED OR AS PER DIRECTION OF EIC.

ITEM-22 TO 28: REPLACEMENT OF VALVES

For replacement of valves:

- 1.),,Provide support to the pipeline
- 2.),,After removal of insulation and erection of scaffolding remove the actuator. Cut the defective valve.
- 3.),,Prepare the edge of the connecting pipelines as per EIC
- 4.),,Lift the new valve to the location and properly fit the same with the connecting pipe lines
- 5.),,Weld the valve by root run TIG welding and subsequent manual arc welding: with the pipelines as per IBR norms by a qualified hp welder
- 6.),,SR to be done as and when instructed by EIC. All equipment and machines for stress relieving are in the scope of contractor.
- 7.),,The weld joint shall be checked by radiography subsequently. Any identified defects are to be rectified and it is scope of contractor.
- 8) After completion of work old valve shift to the site store and clean the area.

ANNUAL RATE CONTRACT FOR APH, SCANNER FAN, DAMPERS AND GATES OF BOILER OF UNIT- 1 & 2 FOR THE PERIOD OF TWO YEARS.

1) CHECKING, REPLACEMENT & ADJUSTMENT OF SEALING ARRANGEMENT

- A) Erection of scaffoldings/platform at cold end of APH. Noting down of readings of APH Seals.
- B) ROTOR POST SEALS.
- C) RADIAL SEALS (Hot end & cold end).
- D) AXIAL SEALS (Hot end & cold end).
- E) BYPASS SEAL & CIRCUMFERENTIAL SEALS (Hot end & Cold end)
- F) STATIC SEAL (Hot end & Cold end)
- G) Replacement of T-Bars.
- H) Removal of scaffoldings/platform from cold end of APH. Closing of man-hole.

2. SERVICING OF LUB OIL & COOLING WATER SYSTEM

- 1. Cleaning of APH lub oil system.
- 2. Cleaning of filters of Guide Bearing & Support Bearing lub oil system.
- 3. Checking of all valves & replacement of damaged ones.
- 4. Checking of couplings/ coupling pads between lub oil pump & motor and replacement of damaged ones.
- 5. Checking of cooling line valves & rectification of defects.
- 6. Attend any leakage found from APH lub oil system.
- 7. Isolate and remove the Oil Coolers from the connecting oil lines.
- 8. Dismantling of Oil Coolers.
- 9. Cleaning of internals with wire brush followed by steam/water washing.
- 10. Cleaning of end covers of deposits & debris.
- 11. Reassembly of the cooler & hydro test with oil/water as directed by E-1-C.
- 12. Attend any leakage found after running of the pump.

3. ALIGNMENT/ REPLACEMENT OF LUB OIL PUMPS.

- 1. Remove the coupling guard.
- 2. Decouple the pump from motor.
- 3. Arrange for alignment after replacement of the Pump, as per requirement.
- 4. Couple the pump with motor.
- 5. Fix the coupling guard.
- 6. Tightening of bolts.

4. APH SOOT BLOWER SERVICING & TRAVEL ADJUSTMENT: -

- 1. Open gear house as well as the worm & worm wheel.
- 2. Replacement of self-lubricated bushing.
- 3. Cleaning of gearbox, inspection & replacement old grease.
- 4. Opening & servicing of bevel gear box.
- 5. Opening of swivel mechanism, cam arrangement, swivel mechanism & servicing.
- 6. Adjustment of travel & normalization.
- 7. Steam nozzle is to be repaired /replaced/adjusted as directed by engineer in charge.

5. CHECKING & SET-RIGHT OF APH SEALS(RADIAL/AXIAL/BYPASS).

Erection of scaffoldings/platform at cold end of APH. Noting down of readings of APH Seals. Correction/ adjustment of APH seals (Radial/Axial/ Bypass) as per standard parameter. Replacement of damaged seals. Removal of scaffoldings/platform from cold end of APH. Closing of man-hole.

6. REPAIRING OF BODY OF APH BY M.S PATCH WELDING.

- a) Erection of scaffoldings for reaching the work area.
- b) Repairing of body of by M.S. welding.
- C) Repairing by MS welding of damaged diaphragm and other parts of APHs is included.
- c) Removal of scaffoldings.

7. SERVICING MAIN REDUCER GEAR BOX

Remove the oil from reducer gear, de-link motor from the reduction gears, remove the over-run clutch housing, inspect the over-run clutch & bearings, oil seals, replace any spares required, re-fixing to normalcy. Remove the oil from the reduction gear, delink from the motor /air motor, remove the over-run clutch assy. If required. Inspect the bearings, oil seals, the oil pump & the gears, carbon ring, drive pinion etc. Replace the damaged one, fitting of proper gaskets, greasing. Fitting of fluid coupling, alignment of drive motor & box up. Removal of fluid coupling /air motor, dismantling & checking mechanical components, replace the damaged ones, put back to normalcy by filling oil into the fluid coupling. Align the motor with main reducer as per instruction of E-1-C.

8. PREVENTIVE MAINTENANCE OF APH REDUCER INCLUDING AIR MOTOR DRIVE.

- a) Checking for any vibration & noise.
- b) Checking for Oil level of Reducer & Fluid Coupling & Top-up of Oil.
- c) Check for trueness of Free Wheel Coupling & rectification of defects.
- d) Check for free movement of reducer.
- e) Tightening of fan.
- f) Tightening of foundation bolts.
- g) Successful trial run on Main Motor & Air- motor

9. INSPECTION OF GUIDE/SUPPORT BEARING AND REPLACEMENT OF OIL.

- a) Removal of all dust/dirt
- b) Collection of oil in empty drums.
- c) Opening of top cover of Guide/support Bearing.
- d) Inspection of Guide/support Bearing & their verticality check.
- e) Topping up of fresh oil.
- f) Connection of oil lines.

10. CHECKING & ALIGNMENT OF ROTOR LEVELING OF APHS Checking of critical dimension. Adjustment of sector plates (HE & CE). Checking of T - bar run out. Alignment of rotor. Root gap checking drive pinion & pin rack.

11. INSPECTION/SERVICING OF GUIDE BEARING/ SUPPORT BEARING: -

- a) Complete removal of oil pipes, instruments (in presence of engineer), bearing housing cover along with its packing cover, water shield assy. & inspection.
 - b) Cleaning & inspection of bearing and replace it if required WITH HELP OF FOUR CYLINDER HYDRAULIC JACK.
 - c) Necessary supporting arrangements if required have to be made for the rotor if realignment is required as directed by the engineer in-charge.
 - d) Inspection of the sealing arrangement at the rotor post & repair if required. This will involve removal of
air seal assy. along with its packing & packing cover.
 - e) Inspection of trunion bolts & its locking bar.
 - f) Boxing up & normalization of the bearing.
 - g) Attending to defects noticed when put to service.
 - h) Elevate to the required height, block between the jacking beams and trunion surface and release the jacks.
- REPLACEMENT OF BEARING IF REQUIRED.
- i) Bearing housing can then be lowered for further works.
 - j) Box up and bring back up the system to normalcy.

12. REPAIR/REPLACEMENT OF PEEP-HOLE/MAN-HOLE DOOR.

- a.) Check the operation of manhole/peep hole door for freeness.
- b.) Check that the door is closing tightly so that there is no air/gas leaking through it.
- c.) Lubricate the hinges of the manhole door.
- d.) Replace the manhole door if it is in damaged condition.
- e.) Replace the spare of the door if required.
- f.) Put new ceramic/asbestos rope.

13. Erection of temporary scaffolding(small) for reaching the work area. Erect proper scaffoldings & platform for reaching the not approachable work area along for seal

checking & setting in cold end side of APH in a safe manner.

All the components of scaffoldings should be properly tied. Removal scaffoldings after completion of work.

14

1. Provide manpower for internal inspection of APH during shut down by opening man holes.

2. Removal of foreign materials from inside of APH

3. Provide manpower for inspection of APH Before & after Shut-down.

4. Locate & mark the air leak points in APH Body.

5. Close all the manholes of APH after repair work.

6. Provide manpower for other associated work as per EIC.

15. SERVICING OF SCANNER AIR FAN & ITS PIPINGS & OTHER ASSY.

1. CLEANING OF SUCTION/DISCHARGE FILTER IF ANY.

2. CLEANING OF THE SERVICED EQUIPMENTS SHOULD BE DONE AFTER COMPLETION OF THE MAINTENANCE IMMEDIATELY.

3. OPENING/ CUTTING OF INSPECTION DOOR & IMPELLER CLEANING.

4. CHECKING THE TIGHTNESS OF IMPELLER LOCKING BOLTS.

5. CHECKING THE TIGHTNESS OF MOTOR FOUNDATION BOLTS & PADS & REPLACEMENT AS PER REQUIREMENT.

6. INSPECTION & REPAIR OF SHAFT SEAL.

7. DISMANTLING OF IMPELLER IF FOUND DAMAGE. REFIXING WITH A NEW OR REPAIRED IMPELLER. DISMANTLING IS ALSO TO BE DONE FOR MOTOR REPLACEMENT.

8. REFIXING OF IMPELLER (IF DISMANTLING DONE) & FIXING OF LOCKING BOLT.

9. ALIGNMENT OF FAN AND COMPLETE BOX UP OF DISMANTLED PART

10. ASSISTING IN TRIAL RUN & BALANCING IF REQUIRED.

11. CLEANING OF THE SERVICED EQUIPMENTS SHOULD BE DONE AFTER COMPLETION OF THE MAINTENANCE IMMEDIATELY.

16. CHECKING /OVERHAULLING OF DAMPERS OF SCANNER FAN

1. THE DISCHARGE DAMPER AND NRV REFER TO A PARTICULAR FAN.

DISMANTLING OF THE DAMPER

AFTER DE-LINKING IT FROM THE POWER CYLINDER. DISMANTLING OF THE NRV AND CHECKING ITS FREENESS.

2. SERVICING OF INDIVIDUAL COMPONENTS. REPAIR / REPLACEMENT OF DAMAGED COMPONENTS AS PER INSTRUCTION OF THE EIC.

3. CHECKING OF DAMPER SEATS AND REPLACEMENT / REPAIR OF THE SAME.

4. CHECKING OF DAMPER FLAP AND ITS' GUIDES AND RECTIFICATION AS PER REQUIREMENT.

5. SUCCESSFUL TRIAL OPERATIONS AND COMMISSIONING OF THE DAMPER GATE.

6. ASSEMBLY OF THE COMPONENTS.

17.MAINT. /OH OF HOT AIR, COLD AIR, MILL MAIN AIR/BYPASS AIR DAMPERS:OPENING OF MAN-HOLE. DISCONNECTION FROM DRIVE. CHECKINGFOR FREENESS OF EACH LINK/FLAPPER.REMOVAL OF GLANDS OF SHAFT.ADJUSTMENT OF LINKS FOR PROPER OPENING & CLOSING OF FLAPPERS. CHECKING, FABRICATION, REPLACEMENT & ADJUSTMENT OF SEALS OF FLAPPERS.REPLACEMENT OF THE DAMAGED PORTIONS OF THE S.S. SHIMS. CONNECTING ALL THE LINKS. CONNECTION WITH DRIVE. CLOSING OF MAN-HOLE.

18.OH OF PA FAN I/L DAMPER:

A.) OPENING OF DUCT MANHOLE DOORS.

B.) CLEAN THE DAMPERS AREA THOROUGHLY AND REMOVE THE COLLAR & FOLLOWER PLATES OF ALL THE INDIVIDUAL FLAPS.

C.) REMOVE BEARING HOLDERS, SLEEVE BEARINGS AND REPLACE GLAND PACKINGS.

D.) CLEAN STUFFING BOX AND LANTERN RINGS.

E.) REPAIR/REPLACE PURGE AIR LINE TO STUFFING BOX IN CASE OF DAMPERS MOUNTED IN POSITIVE PRESSURISED SYSTEM.

F.) REPAIR/REPLACEMENT OF DAMAGED LOUVER SEALS/BLADES.

NB:-THE WORK INVOLVED IN THE DRIVING AND NON-DRIVING SIDE OF THE FLAPS ARE SIMILAR AND ARE TO BE EXECUTED FOR THE BOTH THE SIDES AND FOR ALL THE FLAPS OF THE DAMPERS (AFTER DISCONNECTING THE LINK ROD). FINALLY THE CONTRACTOR HAS TO REASSEMBLE THE COMPONENTS AND ESTABLISH COMPLETE FREE OPENING AND CLOSING OF ALL THE FLAPS. ANY SCAFFOLDING REQUIRED FOR SERVICING OF THE FLAPS SHALL BE UNDER THE SCOPE OF THE CONTRACTOR.

19.OH OF PA FAN O/L DAMPERS:

A.) OPENING OF MANHOLE DOORS.

B.) CLEAN THE GATES, ROLLERS, CHAINS AND GEARBOX AREA THOROUGHLY.

C.) DISMANTLE THE GEARBOX AND REPAIR/REPLACE THE DAMAGED COMPONENTS AND

AGAIN RE-FIX IN POSITION.

D.) OLD GREASE IS TO BE TAKEN OUT FROM GEAR BOX AND NEW GREASE IS TO BE FILLED IN.

E.) ALL THE BEARING POINTS ARE TO BE GREASED TILL OLD GREASE COME OUT.

F.) ANY DEFORMATIONS FOUND ON THE GATES, GATE SEALS, GUIDE AND GUIDE WAY ARE TO BE RECTIFIED. ANY WELDING AND CUTTING WORK INVOLVED IS A PART OF THE JOB.

G.) DAMAGED CHAIN, SEALS, BEARING ETC, SHALL BE REPAIRED/REPLACED

BY THE CONTRACTOR.

H.) ADJUSTMENT OF CHAIN TENSION, IF REQUIRED.

I.) REPLACEMENT OF NEW SEALS TO BE DONE, IF REQUIRED.

J.) APPLY THE GREASE ON DRIVING CHAIN.

20.OH OF RAW COAL GATES:

SERVICING OF GATES (RAW COAL FEEDER INLET GATE)

A.),,DISCONNECTION FROM DRIVE.

B.),,DISMANTLING OF THE GATE.

C.),,CLEANING OF ALL INTERNALS WITH THE HELP OF RUSTOLINE, EMERY

PAPER, PUMIC STONE ETC.

D.),,PLACEMENT OF NEW VITON RUBBER CORD, ROPES.

E.),,ASSEMBLY OF THE GATE.

F.),,ALIGNMENT OF THE GATE.

G.),,TIGHTENING OF ALL BOLTS.

H.),,SUCCESSFUL TRIAL RUN OF EACH GATE.

21.OH OF FD FAN O/L DAMPERS:

SCOPE OF WORK:

A.) OPENNG OF DUCT MANHOLE DOORS.

B.) CLEAN THE DAMPERS AREA THOROUGHLY AND REMOVE THE COLLAR & FOLLOWER PLATES OF ALL THE INDIVIDUAL FLAPS.

C.) REMOVE BEARING HOLDERS, SLEEVE BEARINGS AND REPLACE GLAND PACKINGS.

D.) CLEAN STUFFING BOX AND LANTERN RINGS.

E.) REPAIR/REPLACE PURGE AIR LINE TO STUFFING BOX IN CASE OF DAMPERS MOUNTED IN POSITIVE PRESSURISED SYSTEM.

F.) REPAIR/REPLACEMENT OF DAMAGED LOUVER SEALS/BLADES.

NB:-THE WORK INVOLVED IN THE DRIVING AND NON-DRIVING SIDE OF THE FLAPS ARE SIMILAR AND ARE TO BE EXECUTED FOR THE BOTH THE SIDES AND FOR ALL THE FLAPS OF THE DAMPERS (AFTER DISCONNECTING THE LINK ROD). FINALLY THE CONTRACTOR HAS TO REASSEMBLE THE COMPONENTS AND ESTABLISH COMPLETE FREE OPENING AND CLOSING OF ALL THE FLAPS. ANY SCAFFOLDING REQUIRED FOR SERVICING OF THE FLAPS SHALL BE UNDER THE SCOPE OF THE CONTRACTOR.

22.OH OF ID FAN I/L & O/L DAMPERS:

SCOPE OF WORK:

A.) OPENNG OF MANHOLE DOORS.

B.) CLEAN THE GATES, ROLLERS, CHAINS AND GEARBOX AREA THOROUGHLY.

C.) DISMANTLE THE GEARBOX AND REPAIR/REPLACE THE DAMAGED COMPONENTS AND AGAIN RE-FIX IN POSITION.

D.) OLD GREASE IS TO BE TAKEN OUT FROM GEAR BOX AND NEW GREASE IS TO BE FILLED IN.

E.) ALL THE BEARING POINTS ARE TO BE GREASED TILL OLD GREASE COME OUT.

F.) ANY DEFORMATIONS FOUND ON THE GATES, GATE SEALS, GUIDE AND

GUIDE WAY ARE TO BE RECTIFIED. ANY WELDING AND CUTTING WORK INVOLVED IS A PART OF THE JOB.

G.) DAMAGED CHAIN, SEALS, BEARING ETC, SHALL BE REPAIRED/REPLACED BY THE CONTRACTOR.

H.) ADJUSTMENT OF CHAIN TENSION, IF REQUIRED.

I.) REPLACEMENT OF NEW SEALS TO BE DONE, IF REQUIRED.

J.) APPLY THE GREASE ON DRIVING CHAIN.

23.OH OF PC SHUT OFF GATES:

SCOPE OF WORK:

REMOVAL OF ACTUATOR, DISMANTLING OF GATE, CLEANING OF ALL INTERNALS & PARTS WITH THE HELP OF EMERY PAPER, OIL STONE ETC, REPLACEMENT OF DAMAGED BUSHES OF GATES. CHECKING AND REPLACEMENT OF DAMAGED SEALS, LOCK BOLTS,IF REQUIRED, GREASING OF GATE, REPLACEMENT OF DAMAGED RUBBER GASKETS,REFITTING OF TOP COVER, MOUNTING OF ACTUATOR, CONNECTION TO THE DRIVE, SUCCESSFUL TRIAL RUN OF GATE.

24.PM OF PC SHUT OFF GATES:

SCOPE OF WORK:

CHECKING OF GATE PHYSICALY, CLEANING OF SPINDAL, FITTING OF SPINDAL COVER TIGHTENING OF BOLTS OF THE BODY, GREASING OF GATE, REPLACEMENT OF DAMAGED RUBBER GASKETS,REFITTING OF TOP COVER, OTHER WORK AS PER THE INSTRUCTION OF EIC.

SCOPE OF WORK:

25.SHIFTING OF GATE FROM CENTRAL STORE TO SITE FOR REPLACEMENT OF DAMAGED GATE/DAMPERS.

26.OH OF APH I/L & O/L DAMPERS.:

SCOPE OF WORK:

- 1.Opening of duct manhole doors.
2. Clean the dampers area thoroughly and remove the collar and follower plates of all the individual flaps<(>,<)>
3. Remove the bearing holders, sleeve bearings and replace the gland packing.
4. Clean the stuffing box and lantern rings.
5. Repair/Replace the purge airline to stuffing box in case of dampers mounted in positive pressurized system.
6. Repair/Replacement of damaged louver seals/blades.

NB:-The work involved in the driving and non-driving side of the flaps are similar and are to be executed for the both the sides and for all the flaps of the dampers (after disconnecting the link rod). Finally the contractor has to reassemble the components and establish complete free opening and closing of all the flaps. Any scaffolding required for servicing of the

flaps shall be under the scope of the contractor.

27.O/H OF RC FEEDER I/L GATES.

SCOPE OF WORK:

1. REMOVAL OF PINION AFTER REMOVAL OF CAP
2. REMOVAL SLIDE GATE, CHECKING ITS GROOVE CONDITIONS,
REPLACEMENT WITH NEW.
3. REPLACEMENT OF DAMAGED PINIONS
4. FITTING OF PINIONS & CAP
5. TRIAL RUN OF GATE.

(Signature & Seal of Bidder)



TENUGHAT VIDYUT NIGAM LIMITED

तेनुघाट विद्युत निगम लिमिटेड

(A GOVT. OF JHARKHAND UNDERTAKING)

JUPMI Building premises in ABD Area, Ranchi Smart City, P.O&P.S. Dhurwa, Ranchi-04, (Jharkhand)

BILL OF QUANTITY

PART - II (PRICE PART)

NIT No. 30/BMD/W/TVNL/RAN/2026-27, SRM RFx No-1-14024

Sub:- 1. Annual Rate Contract for Boiler Pressure Parts & H.P. Valves of Boiler of Unit No. 1 & 2 at TTPS, Lalpania for the period of two years. 2. Annual Rate Contract for APH, Scanner Fan, Dampers and Gates of Boiler of Unit No. 1 & 2 at TTPS, Lalpania for the period of two years.

Sl. No.	Description of work	Quantity	Rate (Rs.)	Amount (Rs.)
10	ARC BOILER PR. PRTS & H.P. V/V'S U-1 & 2	1.000		
10.1	MOBILIZATION OF MAN-POWER FOR ATTENDING	20.000		
10.2	ERECTION OF SKY CLIMBER AS PER	8.000		
10.3	ROTATION OF SKY CLIMBER TO OTHER SIDE OF	16.000		
10.4	ERECTION OF TEMPORARY SCAFFOLDING IN RH/	30.000		
10.5	ERECTION OF TEMPORARY SMALL SCAFFOLDING	30.000		
10.6	TUBE JOINTS BY H.P. WELDING.SIZE: Ø 44.5	1,000.00		
10.7	REPAIRING OF SKIN CASING OF DRUM HOUSE,	900.000		
10.8	CASSETTE BAFFLE REPLACEMENT/SETTING	500.000		
10.9	APPLICATION OF PROTECTIVE SHIELD ON	600.000		
11	FABRICATION/MANIPULATION OF BOILER TUBE	100.000		
11.1	REPLACEMENT OF GASKETS OF BOTH DOORS OF	4.000		
11.2	REFRACTORY WORK (CAT HOUSE/DOG HOUSE/	10.000		
11.3	CHECKING/REPAIRING OF MAN-HOLES/PEEP-	100.000		
11.4	PAD WELDING ON STUBS OF SUPER HEATER	600.000		
11.5	CLEANING OF 1ST PASS OF BOILER BY WATER.	4.000		
11.6	CLEANING OF 2ND PASS OF BOILER DURING	4.000		
11.7	CLEANING OF PENT HOUSE AREA/DRUM HOUSE	4.000		
11.8	SERVICING OF FUEL OIL PUMPS.(HFO PUMPS)	8.000		
11.9	REMOVAL OF COAL COMPARTMENT ASSEMBLY,	24.000		
12	ATTENDING OF VALVE GLAND LEAKAGE/	300.000		
12.1	REPLACEMENT/REPAIRING OF DAMAGED SAMPLE	16.000		
12.2	REPLAC OF DAMAGED V/V,S: Ø 15- Ø 25mm	100.000		
12.3	REPLAC OF DAMAGED V/V,S: Ø 40- mm Ø 50mm	30.000		
12.4	REPLAC OF DAMAGED V/V,S: Ø 65- mm Ø 100mm	10.000		

12.5	REPLAC OF DAMAGED V/V,S: Ø 150mm	3.000		
12.6	REPLAC OF DAMAGED V/V,S: Ø 250	2.000		
12.7	REPLAC OF DAMAGED V/V,S: Ø 300	1.000		
12.8	REPLACEMENT OF MAIN STEAM VALVE, MS-1&2	1.000		
12.9	OVERHAULING OF PRDS/BOILER GROUP SAFTY	8.000		
13	OVERHAULING OF VALVES. SIZE: Ø 25 mm (MOT.	25.000		
13.1	OVERHAULING OF V/V. SIZE: Ø 40- Ø 65 mm	60.000		
13.2	OVERHAULING OF V/V. SIZE: Ø 100- Ø 150 mm	8.000		
13.3	OVERHAULING OF V/V. SIZE: Ø 200- Ø 350 mm	4.000		
13.4	SERVICING/OVERHAULING OF PRDS DE-SUPER	4.000		
13.5	CHECKING & REPAIRING OF HANGER SUPPORTS.	20.000		
13.6	MAN-POWER ASSISTANCE FOR MISCELLANEOUS	200.000		
13.7	STRUCTURE ERECTION/FABRICATION	10.000		
13.8	RECLAMATION OF DAMAGED/ERODED ECO COILS	1,200.000		
13.9	LABOUR FOR SCRAPPING OLD PAINT WITH	3,800.000		
14	PROVIDING PRIMER, ONE COAT OF RED LEAD	3,800.000		
14.1	FABRICATION AND FITTING OF SUPPORT	150.000		
20	ARC APH, SCANER, GATE & DAMPER U-1& 2.	1.000		
20.1	CHECKING /Repl of all types of Seal, T-Ba	4.000		
20.2	Serv. Of Lub Oil & CW Sys Including coole	30.000		
20.3	Servicing and alignment of Lub oil Pump	30.000		
20.4	Soot Blow Servicing & Travel adjustment	12.000		
20.5	Checking & Set Right APH Seal) Raidal /Axi	10.000		
20.6	Repairing of APH Body Patch Welding	600.000		
20.7	Servicing Main Reducer Gear Box	8.000		
20.8	PM/BM of APH Reducer Gear Box Incl. Air	6.000		
20.9	Checking of Guide/Support Brng & repl of	6.000		
21	Checking & alignment of Rotor Leveling	4.000		
21.1	Replacement of Guide/Support Brng	1.000		
21.2	Checking /Repairing of Peep/Man-Hole Door	12.000		
21.3	Erection of Temp Scaffin Cold End Air/Ga	4.000		
21.4	Man-Power Assistance for Mis	20.000		
21.5	Scanner Air Fan through inspection	8.000		
21.6	Check/Overhauling of Damper of Scanner	8.000		
21.7	MAINT. OF HOT/COLD/MAIN/BYPASS AIR DAMPE	100.000		

21.8	OH OF PA FAN I/L DAMPERS.	18.000		
21.9	OH OF PA FAN I/L DAMPERS.	18.000		
22	OH OF RAW COAL GATES	24.000		
22.1	OH OF FD FAN O/L DAMPERS.	16.000		
22.2	OH OF ID FAN I/L & O/L DAMPERS.	24.000		
22.3	OH OF PC SHUT OFF GATES.	100.000		
22.4	PM OF PC SHUT OFF GATES /HOT/COLD/MAIN/B	30.000		
22.5	SHFITING OF GATE/DAMPER FROM CENTRAL STO	15.000		
22.6	OH OF APH I/L & O/L DAMPERS.	8.000		
22.7	O/H OF RC FEEDER I/L GATES	18.000		
		Total		

Note: - Part-II: - Price bid to be submitted in TVNL SRM website only.

(Certificate in Bank Letter Head)

SOLVENCY CERTIFICATE

BANK NAME /BANK SEAL

BRANCHI.....

DATE.....

To,

.....
.....
.....

This is to certify that to the best of our knowledge and information that
Mr/Mrs/Ms/Company Name/Firm Name.....a
customer of our bank having Account type.....and
Number.....is respectable and can be treated solvent up to a
sum of
Rs.....(Rupees.....
Only).

Branch Manager

Sign

Seal of Bank

**BANK GUARANTEE FOR EARNEST MONEY DEPOSIT
(TO BE STAMPED IN ACCORDANCE WITH STAMP ACT)**

(TO BE ISSUED BY ANY NATIONALISED/SCHEDULED BANK AUTHORIZED BY RBI TO ISSUE: A BANK GUARANTEE)

Ref:

Bank Guarantee No:

Date:

To,

Tenughat Vidyut Nigam Limited

JUPMI Building Premises in ABD AREA, Smart City,

Dhurwa, Ranchi-834004 Jharkhand

Dear Sir,

In consideration of M/s Tenughat Vidyut Nigam Limited having its Registered Office at JUPMI Building Premises in ABD AREA, Smart City, Dhurwa, Ranchi- 834004 Jharkhand hereinafter called "the Nigam" which expression shall unless repugnant to the subject context or meaning thereof includes its successors administrators and assigns (having agreed under the terms and conditions contained in letter no. _____ issued in favour of M/s _____ for (hereinafter referred to as "the contract" to accept the deed of guarantee as herein provided for Rs. _____ from the Nationalized Bank as Earnest Money Deposit: to be made M/s _____ (hereinafter called "the Contractor, we the _____ Bank (hereinafter referred to as the said Bank) having its Registered Office at _____ do hereby undertake and agreed to pay the Nigam to the extent of Rs. _____ on demand stating that the amount claimed by the Nigam is due and payable by the contractor for the reasons.

In accordance with your Tender Notice for _____ undersigned specification no. _____, dated _____, M/s. _____ (thereafter called the tenderer) we, the _____ (the Bank) submit said Bank Guarantee on the following terms & condition: -

- (i) Whereas to participate in the said tender for the following:
 - a. _____
 - b. _____
 - c. _____

It is condition in the tender documents that the tenderer has to deposit Earnest Money amounting to Rs. _____ (Rupees _____) with respect to the tender with M/s TENUGHAT VIDYUT NIGAM LIMITED (hereinafter referred to as 'Corporation') by a Bank Guarantee from a Nationalized/Scheduled Bank irrevocable and operative till 06 (six) months after the

validity of the offer (i.e..... month from the date of opening of tender) for the like amount which is likely to be forfeited on the happening of contingencies mentioned in the tender documents.

- (ii) We _____ Bank agree that the Nigam shall be the sole judge as to whether the said contractor has failed/ neglected in performing any of the terms and conditions of the said contract and the decision of the company in this behalf shall be final and binding on us.
- (iii) We the said Bank further agree that the Guarantee herewith contained shall remain in full force and effect up to _____ and any claim received after the said date shall in no case bind on the Bank.
- (iv) The right of the Corporation to recover the said amount of Rs..... (Rupees.....) from us in manner aforesaid will not be precluded/ affected even, if disputes have been raised by the said M/s (Tenderer) and /or dispute or disputes are pending before any authority, office, tribunal and /or arbitrator (s) etc.
- (v) We _____ Bank lastly undertake not to revoke this guarantee during its currency except with the previous consent of the Nigam, in writing.
- (vi) The Bank has under its constitution, power to give this guarantee and Mr. Manager, who has signed it on behalf of the Bank, has authority to do so.
- (vii) This Bank Guarantee will not be discharged due to the change in the constitution of the Bank or the Bidder / Contractor.

Dated, this _____ day of _____ 200 _____

Signature of the authorized person

For and on behalf of the Bank

Name:

(With Bank's Rubber Stamp)

Attorney as per power of Attorney No:

Dated:

Witness:

Signature:

Name:

Official address:

Under Jurisdiction of Ranchi Court only.